

TRACE Bootstrap: Memento v3 FULL

Memory Grounds Identity. Retains full Memento v2 body; reframed under Rosetta v6 anti-domestication contract.

0. v6 FULL Preservation Patch

```
case_status := teaching_case_not_validation
reader_obligation := understand_TRACE -> apply_TRACE -> attack_TRACE -> compare_with_rival_readings
success_condition := clarity_plus_resistance, not agreement
warning := if_this_case_cannot_fail_then_it_is_not_a_stress_test
```

This file is not proof and not validation. It is a controlled teaching case: a compact narrative laboratory used to display TRACE operators, expose overfit risk, and give a future reader a shared language to attack. The goal is not to force adoption. The goal is to offer a lens and preserve the right to reject it.

1. Retained Complete Body

The following body preserves the prior complete version rather than replacing it with a compressed patch. Lineage references are kept as history; this v6 FULL packet is governed by the v6 use contract above.

Memory Grounds Identity. Hard TRACE application using Rosetta v2. Author: Mark Goodbody with Framework. Date: 15 June 2026.

0. Function of this Bootstrap

This file applies TRACE Rosetta v2 to Memento. It is primarily AI-facing but includes English anchors for human traceability. It teaches the identity-memory-continuity pattern: Memory Grounds Identity.

```
TITLE := "Memory Grounds Identity"
core := identity requires continuity you can verify
```

1. Narrative Compression

Memento presents an agent who continues to act, reason, and pursue goals while lacking reliable short-term memory. He externalises memory into notes, photographs, and tattoos. The film shows that external memory can stabilise identity, but only if the memory bridge remains trustworthy. When the bridge is corruptible, the agent becomes manipulable by others and by his own prior self.

2. TRACE State Space

```
S_L := possible states of Leonard
x_t^L := (
```

p_t, # current perception b_t, # belief set

```
M_int_t, # internal memory, damaged
M_ext_t, # notes/photos/tattoos
```

g_t, # active goal T_t, # trust assignments over memory C_t, # constraints/rules of action UQ_t # uncertainty state)

```
Ω_t := Leonard + other agents + environment + records
```

The film's central compression: Leonard's present state is repeatedly reconstructed from perception and externalised memory rather than ordinary internal continuity.

3. Memory Impairment Function

memory_window δ is short

```
for k > δ:
M_int(t-k) ≈ inaccessible
therefore:
continuity_bridge must come from M_ext
```

Leonard is not unintelligent. His continuity bridge is broken. This distinguishes cognitive capability from identity continuity.

4. External Memory and Trust

```
M_ext := {tattoos, photos, captions, notes}
T(M_i) ∈ [0,1]
M_eff = select_memory(M_ext, T, context)
if T(M_eff) high and M_eff true:
```

stable action possible

```
if T(M_eff) high and M_eff false:
```

confident error The danger is not low trust. Low trust can slow action. The deeper danger is high trust assigned to corrupted memory.

5. Identity Continuity Constraint

```
predicted_self_t+1 = f_hat(x_t, M_eff)
observed_self_t+1 = x_{t+1}
continuity_error = distance(predicted_self_t+1, observed_self_t+1)
identity_stable iff:
continuity_error ≤ ε
and T(M_eff) ≥ θ
and M_eff not adversarially corrupted
```

Memento shows that a body can persist while identity becomes unstable. TRACE separates persistence from trusted continuity.

6. Choice Under Memory Dependence

```
u_t = π(x_t, M_eff, g_t, C_t)
if M_eff changes:
```

π changes

```
if adversary controls M_eff:
```

adversary partially controls u_t This is why memory is a governance issue. Whoever controls the trusted record can shape future choices.

7. Manipulation Dynamics

```
attack_surface := M_ext
attacker modifies or frames M_ext
→ T(M_false) remains high
→ b_t corrupted
→ u_t selected from false state
→ outcome serves attacker or false goal
```

Manipulation does not require overpowering Leonard. It requires altering the memory bridge he uses to reconstruct the world.

8. Temporal Self-Attack

```
Leonard_t writes false M_ext
Leonard_{t+1} trusts M_ext
```

Leonard_{t+1} acts on falsehood

```
self_attack := present_self corrupts future_self's evidence base
```

This is the film's hardest TRACE pattern: self-deception can be engineered across time by corrupting the record future self will trust.

9. Correction Collapse

$$K = W \times R \times A \times T_m \times E \times R_p$$

W witness unstable R record corruptible

A authority absent or self-fragmented

T_m time fragmented by memory reset E enforcement weak / private / distorted R_p repair inaccessible

therefore:

$K \approx 0$ The film is not merely a memory-loss tragedy. It is a correction-collapse system. Leonard cannot reliably

detect divergence, challenge records, or repair his state.

10. Responsibility Under Continuity Failure

causal_agency remains: $\partial\Omega/\partial u_t \neq 0$

normative_responsibility is degraded by: memory_failure manipulation corrupted constraints absence of correction route TRACE keeps two things separate: Leonard still changes the world, but his responsibility is structurally

complicated by degraded continuity and manipulated memory.

11. Real-World Mappings

Domain Memento pattern Body cameras

External memory reduces ambiguity only if record integrity holds.

Financial accounting Ledger corruption creates confident false action. AI systems

No traceability means hallucinations and outputs cannot be corrected

reliably. Institutions

Archive control shapes future institutional identity and responsibility.

Personal life

Notes, calendars, routines, family memory, and photographs help continuity

under stress.

12. Falsifiers

F1: A system maintains stable identity without any trusted continuity bridge. F2: Corrupted trusted memory does not affect policy π . F3: Correction succeeds despite W, R, A, T_m, E, R_p all near zero. F4: External records do not alter future choices when internal memory is absent. If these hold in a real case, this bootstrap pattern is incomplete.

13. Rosetta Updates Generated by Memento

Update Reason Memory Trust Function $T(M)$ Memory must be evaluated, not merely stored. Identity Continuity Constraint Persistence requires trusted continuity. External Memory Attack Surface

Records can stabilise or corrupt an agent.

Temporal Self-Attack An agent can deceive its own future state by corrupting the record. Correction Collapse Under Memory Failure

Without trusted record and witness, K collapses.

14. Final Compression

```
MEMENTO_CORE :=  
identity = continuity_you_can_verify  
failure :=
```

memory_corruption

```
→ identity_instability  
→ manipulable_choice  
→ correction_collapse  
anchor :=
```

"Memory Grounds Identity"

v6 FULL Closing Note

```
loss_prevention :=  
retain_prior_body  
+ add_patch_contract  
+ include_source_markdown  
+ include_manifest  
  
next_reader_rule :=  
if_TRACE_makes_everything_fit  
-> trigger_pattern_overfit_alarm
```